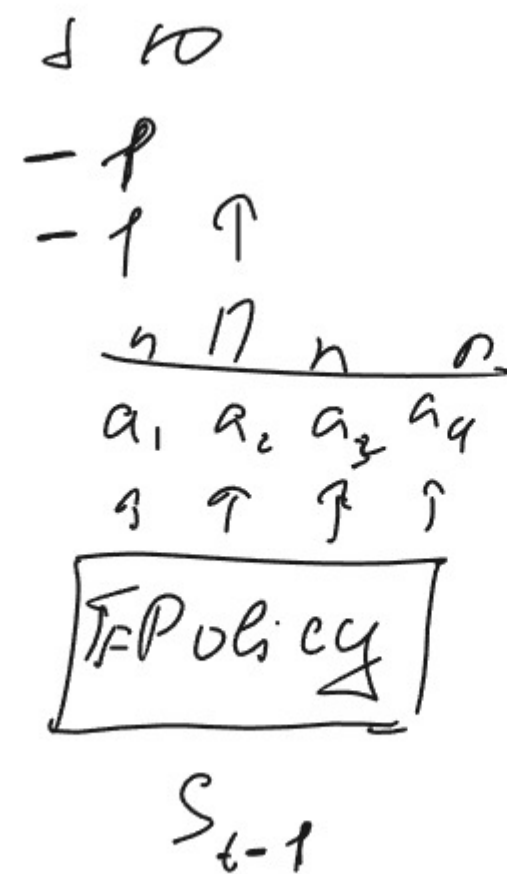
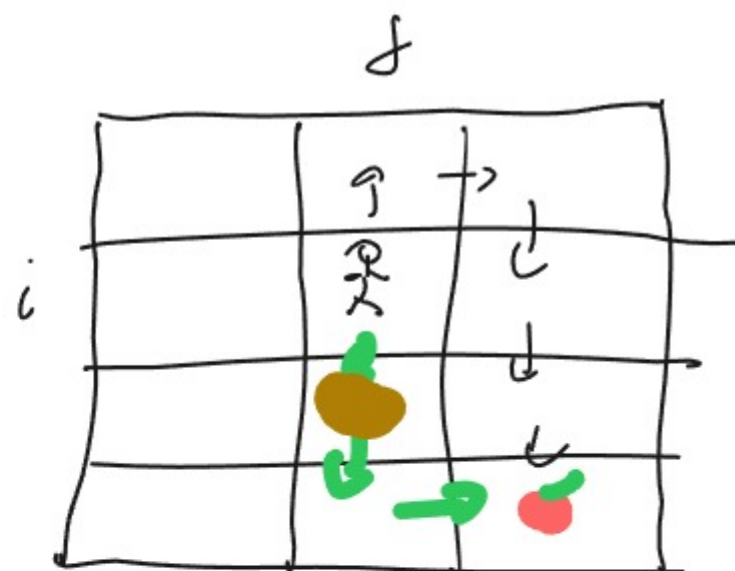


Intro to RL

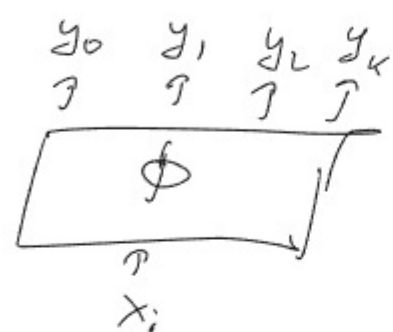


PLHF

PPO

π = model = policy

$$\text{objective}(\phi) = E_{(x,y) \sim D_{\pi^{\text{RL}}}} \left[\underbrace{r_\theta(x,y)}_1 - \underbrace{\beta \log(\pi_\phi^{\text{RL}}(y|x)/\pi^{\text{SFT}}(y|x))}_2 \right] \rightarrow \max$$



$$1: \underline{J}_{x_i} = \sum_{j=1}^K P(y_j | x_i; \phi) \underline{z_\theta(x_i, y_j)} =$$

$$= \sum_{j=1}^K \pi_\phi(y_j | x_i) z_\theta(x_i, y_j) = E_y z_\theta(x_i, y_j)$$

$$J = \sum_{i=1}^N E_y z_\theta(x_i, y_j) \approx \frac{1}{S} \sum_{s=1}^S z_\theta(x_i, y_j)$$

$$\nabla_\phi J = \nabla_\phi \sum_{i=1}^N J_{x_i} = \sum_{i=1}^N \sum_{j=1}^K \nabla_\phi \pi_\phi(y_j | x_i) z_\theta(x_i, y_j) =$$

$$= \sum_{i=1}^N \sum_{j=1}^K \pi_\phi(y_j | x_i) \nabla_\phi \log \pi_\phi(y_j | x_i) \cdot z_\theta(x_i, y_j) =$$

$$= E_{x,y \sim \pi_\phi} (z_\theta(x, y) \cdot \nabla_\phi \log \pi_\phi(y | x))$$

$$\nabla \text{loss} := \underbrace{-\text{obj}}_{\text{loss}} \cdot \underbrace{\nabla \log \text{prob}}_{\text{grad}}$$

t_0, t_1, t_L

$$P(\underset{||}{y_d}) = P(d_0 | \dots) \cdot P(d_1 | d_0) \cdot P(d_2 | d_0, d_1) = \\ = \prod_{i=0}^{\text{seq. len}} P(t_i | t_{<i})$$

Log-derivative
Zick

$$\log' f(x) = \frac{1}{f(x)} f'(x) \Rightarrow$$

$$\Rightarrow f'(x) = f(x) \log' f(x)$$

2.

$$\text{objective}(\phi) = E_{(x,y) \sim D_{\pi_{\phi}^{\text{RL}}}} \left[r_{\theta}(x, y) - \underbrace{\beta \log \left(\pi_{\phi}^{\text{RL}}(y | x) / \pi^{\text{SFT}}(y | x) \right)}_{\text{KL Divergence}} \right]$$

$$\sum P_{\phi}(y|x) \log \frac{\pi^{\text{RL}}(y|x)}{\pi^{\text{SFT}}(y|x)} \approx$$

$$\approx \left(\sum_{i=1}^K P_i \log P_i - P_i \cdot \log (P_i) \right)$$